L'INFECTION NÉONATALE BACTÉRIENNE

Lala N'drainy SIDIBE

Pédiatre

Enseignant chercheur

CONTEXTE

EPIDEMIOLOGIE

PROBLEMATIQUES

PISTE DE SOLUTION

CONCLUSION

Définition

- Manifestations cliniques et paracliniques
- Envahissement de l’organisme par des microorganismes: Bactéries
- Dans la période néonatale: 0 - 28 jours

Définition

J0 H72

Infections précoces

ou

H72 J28

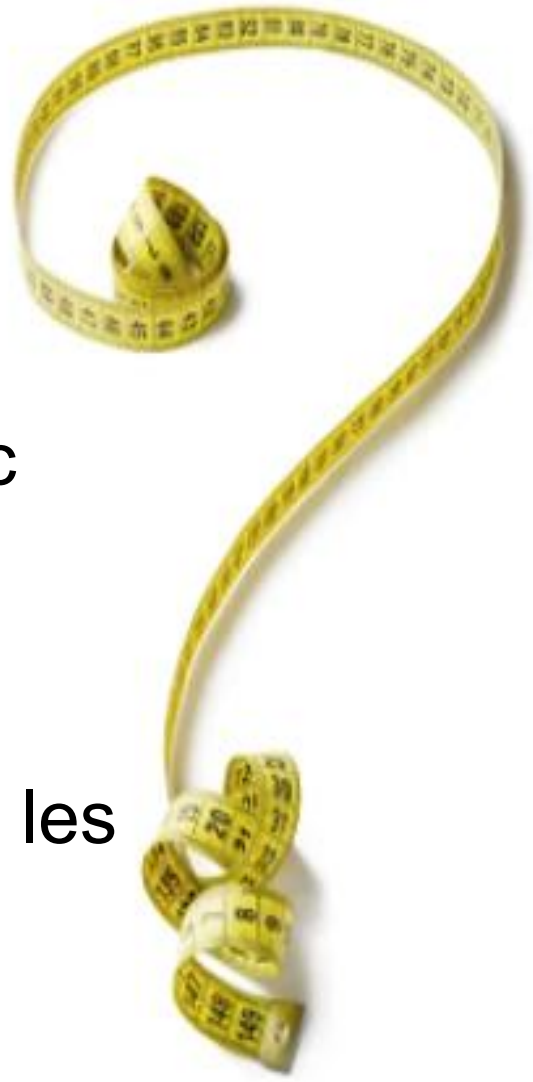
Infections tardives

CONTEXTE

- Afrique subsaharienne 57 % des décès chez les < 5 ans
- Taux de mortalité néonatale le plus élevé au monde avec

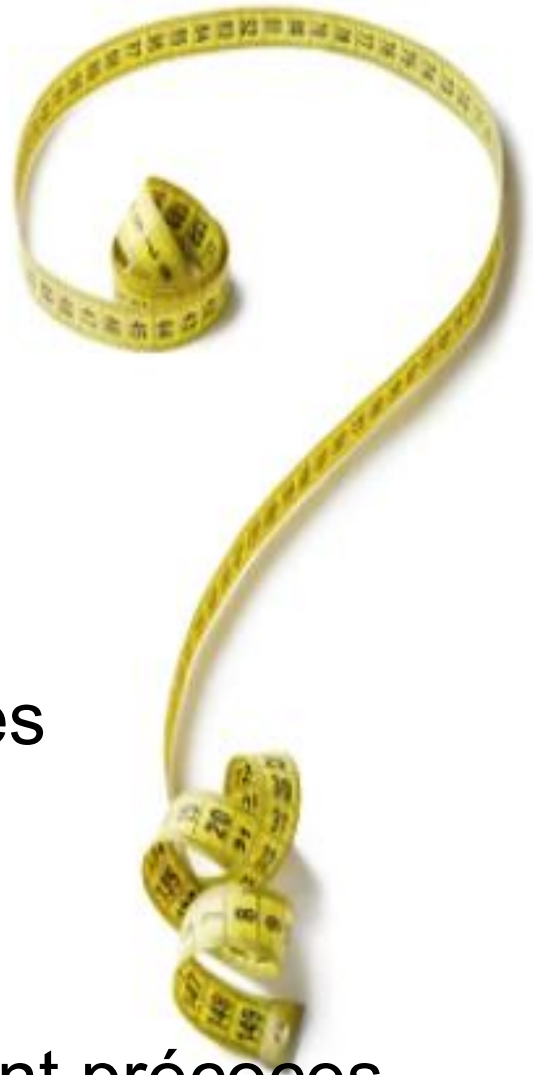
  27 décès pour 1000 naissances
- L’ INN fait partie des 5 principales causes de décès chez les

  nouveau-nés



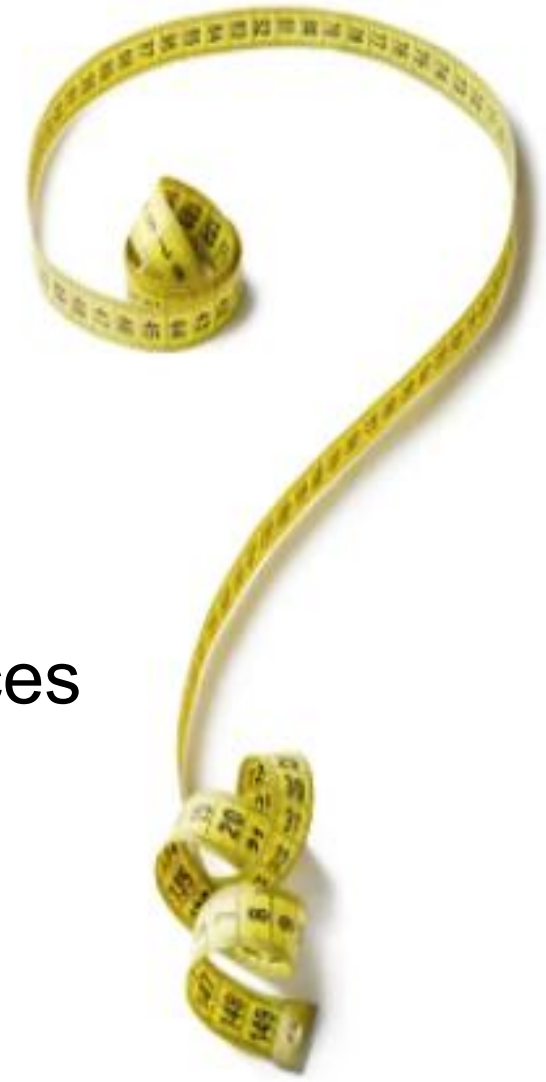
CONTEXTE

- Risque d’infection bactérienne plus élevé
  - immaturité immunitaire /plus susceptible aux IMF
    - Innée: humorale, cellulaire
    - Spécifique: Ac spécifiques, barrières physiologiques
  - Risque plus élevé chez les prématurés
- Gravité accrue en l’absence de diagnostic et de traitement précoces



CONTEXTE

- Causes insuffisance dans la prise en charge
- Cible fixée par ODD d'ici 2030 ne sera pas atteint
  - éliminer les décès évitables de nouveau-nés
  - ramener la mortalité néonatale à 12 pour ‰ naissances vivantes au plus



EPIDEMIOLOGIE

- **Prévalence annuelle mondiale**

  5 millions de cas [1,2]

- **Incidence**: 28,2 à 39,3 *vs* 0,3 à 0,8 pour 1 000 naissances vivantes / LMIC *vs* HIC [3]

- **Mortalité**:
  - environ 800 000 décès /an [1,2]
  - 25 % de tous les décès néonatals dans le monde [3]
  - 18 % contre 0 à 8 % / LMIC *vs* HIC [3]

EPIDEMIOLOGIE

- Au Mali:
  - Incidence hospitalière 33,8% [4]
  - INBP confirmées par l'hémoculture 11,04% [4]
  - 3ème cause d'hospitalisation et de mortalité en néonatologie [5]
  - Mortalité néonatale globale 29 ‰ [6]

EPIDEMIOLOGIE

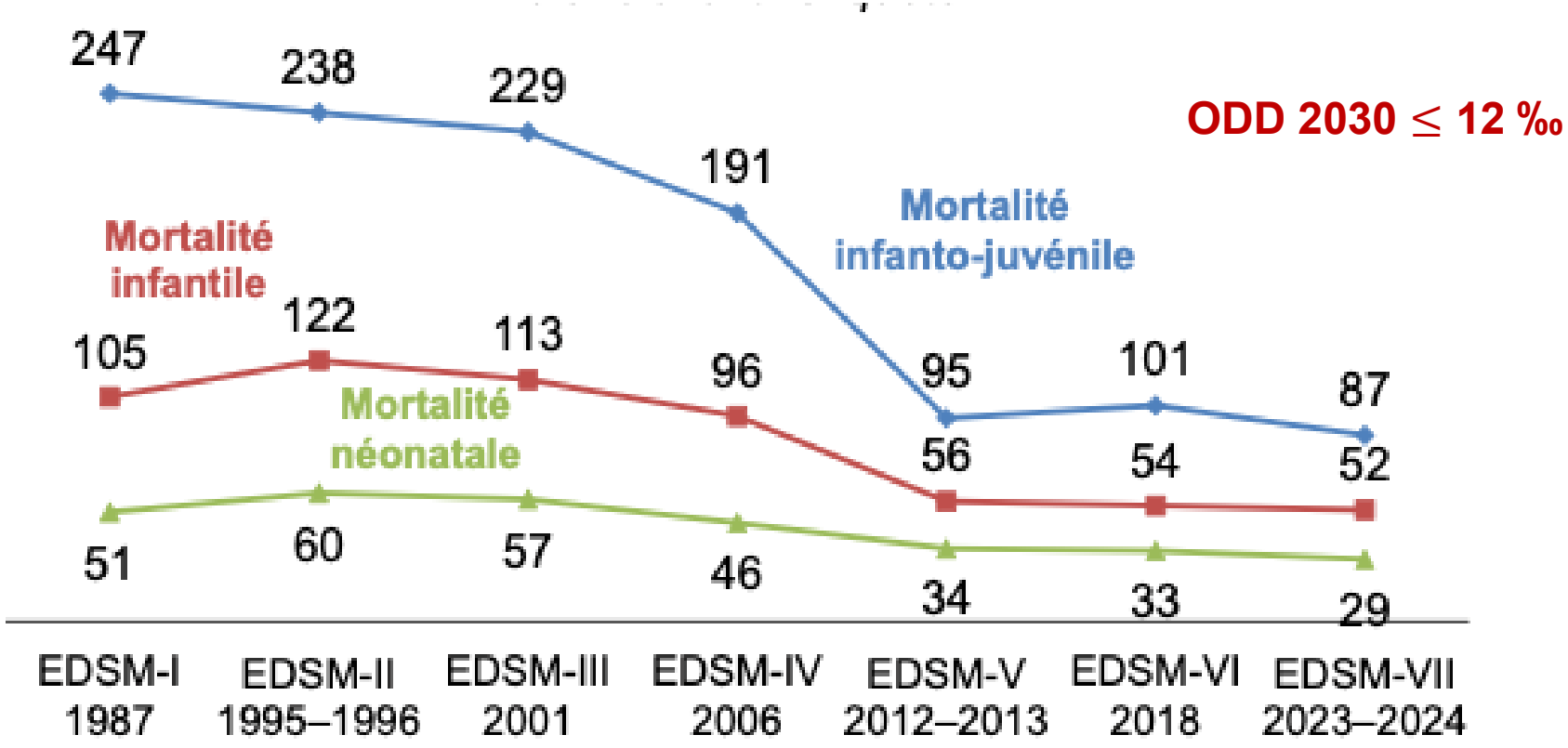


Fig 1: Tendances de la mortalité des enfants de moins de cinq ans

EPIDEMIOLOGIE

Lancet Glob Health 2018; 6: e1297-308

Tableau 1: Les causes de décès néonatal en Asie du Sud et en Afrique Subsaharienne

	Neonatal death	Verbal autopsy done	Cause determined	Preterm birth complications	Perinatal asphyxia	Severe neonatal infection*	Neonatal diarrhoea	Neonatal tetanus	Congenital malformation	Accident or injuries	Other perinatal causes†
Bangladesh	995	971 (98%)	933	146 (16%)	342 (37%)	431 (46%)	0	4 (0%)	7 (1%)	1 (0%)	2 (0%)
India											
Haryana	1399	1301 (93%)	1206	306 (25%)	475 (39%)	321 (27%)	5 (<1%)	2 (0%)	87 (7%)	4 (0%)	6 (0%)
Uttar Pradesh	1575	1481 (94%)	1375	178 (13%)	531 (39%)	531 (39%)	11 (1%)	12 (1%)	54 (4%)	10 (1%)	48 (3%)
Pakistan											
Matiari	1269	1221 (96%)	1167	226 (19%)	497 (43%)	377 (32%)	6 (1%)	2 (<1%)	42 (4%)	0	17 (1%)
Karachi	862	795 (92%)	763	173 (23%)	307 (40%)	224 (29%)	8 (1%)	2 (<1%)	38 (5%)	3 (<1%)	8 (1%)
Democratic Republic of the Congo	173	108 (62%)	111	28 (25%)	23 (21%)	56 (50%)	0	1 (1%)	0	0	3 (3%)
Ghana	687	598 (87%)	578	169 (29%)	210 (36%)	161 (28%)	0	4 (1%)	16 (3%)	3 (1%)	15 (3%)
Kenya	397	271 (68%)	260	102 (39%)	82 (32%)	72 (28%)	1	0	3 (1%)	0	0
Tanzania											
Ifakara	221	168 (76%)	159	22 (14%)	62 (39%)	72 (45%)	0	0	2 (1%)	0	1 (1%)
Pemba	303	276 (91%)	253	33 (13%)	119 (47%)	79 (31%)	0	0	9 (4%)	1 (0%)	12 (5%)
Zambia	363	273 (75%)	261	72 (28%)	70 (27%)	97 (37%)	3 (1%)	2 (1%)	10 (4%)	1 (0%)	6 (2%)
South Asia	6100	5769 (95%)	5444	19% (18–20)‡	40% (39–42)‡	35% (34–36)‡	<1%	<1%	3% (3–4)‡	<1%	1% (1–2)‡
Sub-Saharan Africa	2144	1694 (79%)	1622	24% (22–26)‡	34% (32–36)‡	37% (34–39)‡	<1%	<1%	2% (2–3)‡	<1%	2% (2–3)‡

*Neonatal pneumonia, sepsis, or meningitis. †Other perinatal cause or other conditions originating in the perinatal period. ‡Overall proportion (95% CI) was obtained from meta-analysis.

Table 3: **Causes of neonatal deaths, by site**

EPIDEMIOLOGIE

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EPIDEMIOLOGIE

Uduak Okomo PhD [a], Edem N K Akpalu MD [b], Kirsty Le Doare PhD [c],

Aetiology of invasive bacterial infection and antimicrobial resistance in neonates in sub-Saharan Africa: a systematic review and meta-analysis in line with the STROBE-NI reporting guidelines

The Lancet Infectious Diseases
Volume 19, Issue 11, November 2019, Pages 1219-1234

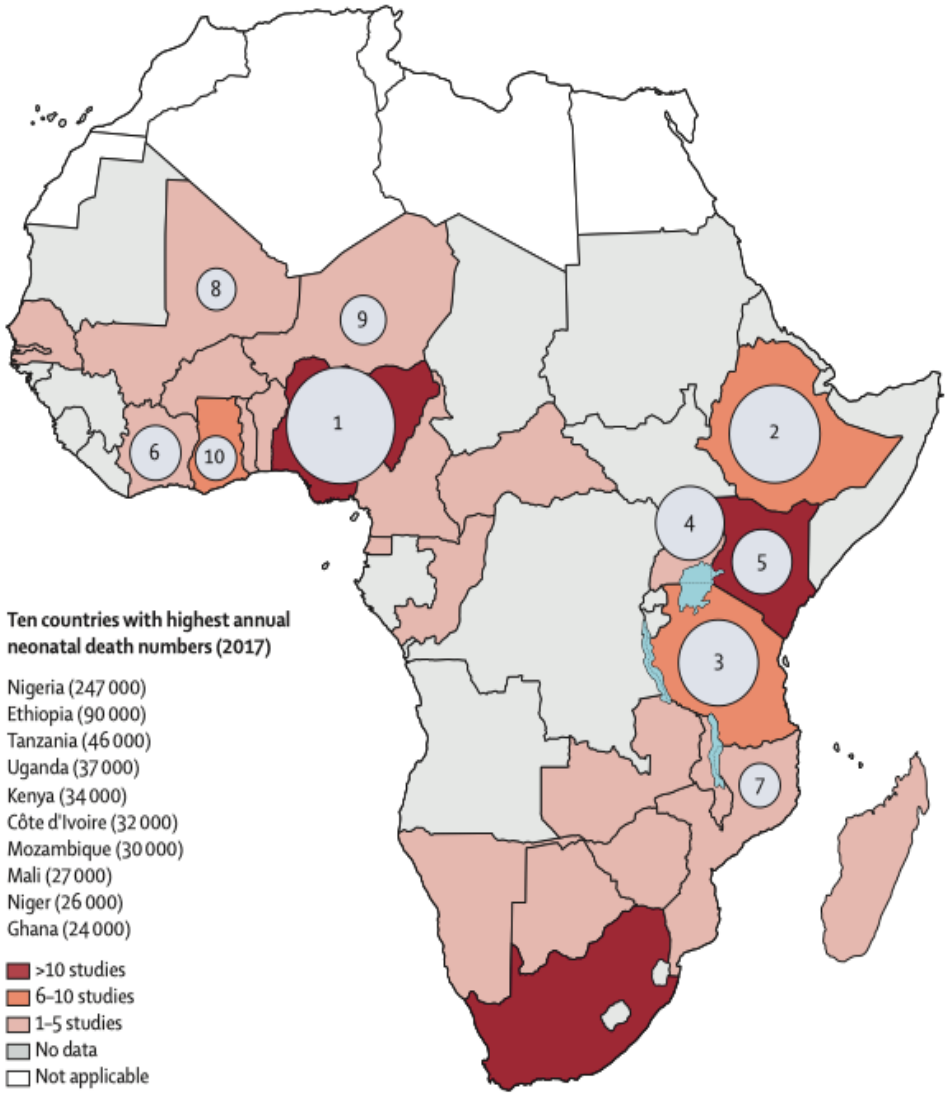

Fig 1: Geographical representation of included studies and variation between in the number of studies per country from 1980 to 2018

EPIDEMIOLOGIE

Germes

	1980–2007		2008–18	
	Number of isolates	Proportion (95% CI)	Number of isolates	Proportion (95% CI)
Bacteraemia or sepsis				
Gram-positive				
Staphylococcus aureus	912	0·25 (0·19–0·31)	2080	0·25 (0·21–0·29)
Streptococcus pyogenes	75	0·04 (0·02–0·08)	117	0·04 (0·02–0·07)
Group B streptococci	213	0·07 (0·03–0·12)	342	0·06 (0·03–0·10)
Group D streptococci or enterococcus	139	0·05 (0·03–0·07)	449	0·05 (0·04–0·07)
Streptococcus pneumoniae	72	0·04 (0·02–0·08)	114	0·02 (0·01–0·04)
Viridians streptococci	7	0·01 (0–0·05)	71	0·03 (0·01–0·05)
Other *Streptococcus* species	63	0·03 (0·01–0·05)	209	0·05 (0·03–0·07)
Other or unspecified Gram-positives	86	0·04 (0·01–0·08)	155	0·06 (0·03–0·09)
Gram-negative				
Klebsiella species	644	0·15 (0·11–0·20)	1730	0·21 (0·16–0·27)
Escherichia coli	377	0·10 (0·08–0·13)	856	0·10 (0·08–0·13)
Pseudomonas species	146	0·04 (0·02–0·05)	189	0·03 (0·02–0·04)
Enterobacter species	270	0·08 (0·03–0·13)	263	0·04 (0·03–0·05)
Serratia species	0	..	129	0·03 (0·01–0·07)
Proteus species	54	0·02 (0·01–0·04)	126	0·03 (0·02–0·04)
Salmonella species	162	0·03 (0·02–0·05)	176	0·04 (0·02–0·06)
Citrobacter species	61	0·04 (0·01–0·07)	122	0·02 (0·02–0·04)
Haemophilus influenzae	11	0·01 (0–0·02)	10	0·01 (0–0·03)
Neisseria meningitidis	0	..	17	0·03 (0–0·08)
Acinetobacter species	94	0·05 (0·02–0·07)	299	0·05 (0·03–0·07)
Other or unspecified Gram-negatives	522	0·20 (0·14–0·27)	508	0·10 (0·06–0·14)
Other pathogens	14	0·05 (0·02–0·07)	9	0·02 (0·01–0·04)

Meningitis				
Gram-positive				
S aureus	77	0·18 (0·07–0·32)	92	0·12 (0·03–0·25)
S pyogenes	10	0·01 (0–0·03)	8	0·07 (0·03–0·12)
Group B streptococcus	297	0·26 (0·18–0·35)	416	0·24 (0·16–0·33)
Group D streptococcus or enterococcus	8	0·03 (0–0·07)	68	0·03 (0·01–0·06)
S pneumoniae	210	0·15 (0·11–0·21)	157	0·17 (0·09–0·26)
Viridians streptococci	0	..	13	0·01 (0–0·03)
Other *Streptococcus* species	36	0·06 (0·02–0·11)	37	0·03 (0·02–0·04)
Other or unspecified	23	0·04 (0·02–0·07)	12	0 (0–0·01)
Gram-negative				
Klebsiella species	150	0·15 (0·09–0·21)	39	0·10 (0·04–0·18)
E coli	170	0·15 (0·10–0·20)	45	0·11 (0·06–0·18)
Pseudomonas species	21	0·04 (0·02–0·08)	6	0·03 (0–0·08)
Enterobacter species	29	0·07 (0·03–0·13)	15	0·06 (0·02–0·11)
Serratia species	11	0·05 (0·02–0·10)	3	0·08 (0·01–0·20)
Proteus species	19	0·03 (0·01–0·05)	2	0·01 (0–0·06)
Salmonella species	68	0·06 (0·03–0·10)	19	0·08 (0·05–0·13)
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H influenzae	36	0·04 (0·02–0·07)	10	0·01 (0–0·04)
N meningitidis	25	0·02 (0–0·08)	20	0·04 (0–0·10)
Acinetobacter species	9	0·04 (0·01–0·08)	9	0·10 (0·04–0·17)
Other or unspecified Gram-negatives	114	0·11 (0·07–0·15)	394	0·12 (0·02–0·25)
Other pathogens	80	0·24 (0·17–0·32)	11	0·01 (0–0·02)

EPIDEMIOLOGIE

Germes

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Group B streptococci	213	0·07 (0·03–0·12)	342	0·06 (0·03–0·10)
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Streptococcus pneumoniae	72	0·04 (0·02–0·08)	114	0·02 (0·01–0·04)
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E coli	170	0·15 (0·10–0·20)	45	0·11 (0·06–0·18)
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Group B streptococci	213	0·07 (0·03–0·12)	342	0·06 (0·03–0·10)
Group D streptococci or enterococcus	139	0·05 (0·03–0·07)	449	0·05 (0·04–0·07)
Streptococcus pneumoniae	72	0·04 (0·02–0·08)	114	0·02 (0·01–0·04)
Viridians streptococci	7	0·01 (0–0·05)	71	0·03 (0·01–0·05)
Other *Streptococcus* species	63	0·03 (0·01–0·05)	209	0·05 (0·03–0·07)
Other or unspecified Gram-positives	86	0·04 (0·01–0·08)	155	0·06 (0·03–0·09)
Gram-negative				
Klebsiella species	644	0·15 (0·11–0·20)	1730	0·21 (0·16–0·27)
Escherichia coli	377	0·10 (0·08–0·13)	856	0·10 (0·08–0·13)
Pseudomonas species	146	0·04 (0·02–0·05)	189	0·03 (0·02–0·04)
Enterobacter species	270	0·08 (0·03–0·13)	263	0·04 (0·03–0·05)
Serratia species	0	..	129	0·03 (0·01–0·07)
Proteus species	54	0·02 (0·01–0·04)	126	0·03 (0·02–0·04)
Salmonella species	162	0·03 (0·02–0·05)	176	0·04 (0·02–0·06)
Citrobacter species	61	0·04 (0·01–0·07)	122	0·02 (0·02–0·04)
Haemophilus influenzae	11	0·01 (0–0·02)	10	0·01 (0–0·03)
Neisseria meningitidis	0	..	17	0·03 (0–0·08)
Acinetobacter species	94	0·05 (0·02–0·07)	299	0·05 (0·03–0·07)
Other or unspecified Gram-negatives	522	0·20 (0·14–0·27)	508	0·10 (0·06–0·14)
Other pathogens	14	0·05 (0·02–0·07)	9	0·02 (0·01–0·04)

Meningitis				
Gram-positive				
S aureus	77	0·18 (0·07–0·32)	92	0·12 (0·03–0·25)
S pyogenes	10	0·01 (0–0·03)	8	0·07 (0·03–0·12)
Group B streptococcus	297	0·26 (0·18–0·35)	416	0·24 (0·16–0·33)
Group D streptococcus or enterococcus	8	0·03 (0–0·07)	68	0·03 (0·01–0·06)
S pneumoniae	210	0·15 (0·11–0·21)	157	0·17 (0·09–0·26)
Viridians streptococci	0	..	13	0·01 (0–0·03)
Other *Streptococcus* species	36	0·06 (0·02–0·11)	37	0·03 (0·02–0·04)
Other or unspecified	23	0·04 (0·02–0·07)	12	0 (0–0·01)
Gram-negative				
Klebsiella species	150	0·15 (0·09–0·21)	39	0·10 (0·04–0·18)
E coli	170	0·15 (0·10–0·20)	45	0·11 (0·06–0·18)
Pseudomonas species	21	0·04 (0·02–0·08)	6	0·03 (0–0·08)
Enterobacter species	29	0·07 (0·03–0·13)	15	0·06 (0·02–0·11)
Serratia species	11	0·05 (0·02–0·10)	3	0·08 (0·01–0·20)
Proteus species	19	0·03 (0·01–0·05)	2	0·01 (0–0·06)
Salmonella species	68	0·06 (0·03–0·10)	19	0·08 (0·05–0·13)
Citrobacter species	12	0·07 (0·02–0·14)	3	0·04 (0–0·11)
H influenzae	36	0·04 (0·02–0·07)	10	0·01 (0–0·04)
N meningitidis	25	0·02 (0–0·08)	20	0·04 (0–0·10)
Acinetobacter species	9	0·04 (0·01–0·08)	9	0·10 (0·04–0·17)
Other or unspecified Gram-negatives	114	0·11 (0·07–0·15)	394	0·12 (0·02–0·25)
Other pathogens	80	0·24 (0·17–0·32)	11	0·01 (0–0·02)

EPIDEMIOLOGIE

Germes

	1980–2007		2008–18	
	Number of isolates	Proportion (95% CI)	Number of isolates	Proportion (95% CI)
Bacteraemia or sepsis				
Gram-positive				
Staphylococcus aureus	912	0·25 (0·19–0·31)	2080	0·25 (0·21–0·29)
Streptococcus pyogenes	75	0·04 (0·02–0·08)	117	0·04 (0·02–0·07)
Group B streptococci	213	0·07 (0·03–0·12)	342	0·06 (0·03–0·10)
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Viridians streptococci	7	0·01 (0–0·05)	71	0·03 (0·01–0·05)
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Gram-negative				
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Meningitis				
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S pneumoniae	210	0·15 (0·11–0·21)	157	0·17 (0·09–0·26)
Viridians streptococci	0	..	13	0·01 (0–0·03)
Other *Streptococcus* species	36	0·06 (0·02–0·11)	37	0·03 (0·02–0·04)
Other or unspecified	23	0·04 (0·02–0·07)	12	0 (0–0·01)
Gram-negative				
Klebsiella species	150	0·15 (0·09–0·21)	39	0·10 (0·04–0·18)
E coli	170	0·15 (0·10–0·20)	45	0·11 (0·06–0·18)
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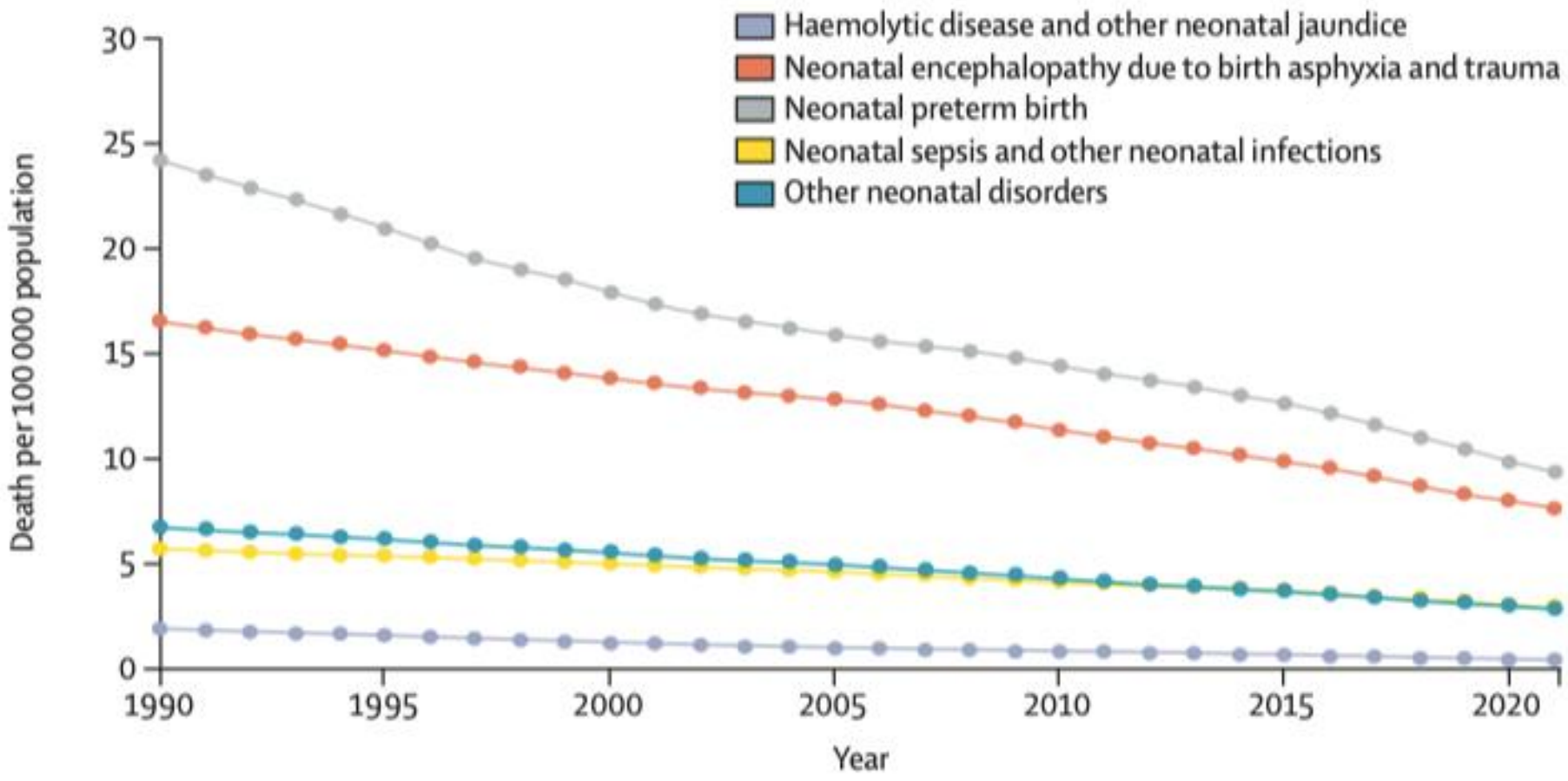
H influenzae	36	0·04 (0·02–0·07)	10	0·01 (0–0·04)
N meningitidis	25	0·02 (0–0·08)	20	0·04 (0–0·10)
Acinetobacter species	9	0·04 (0·01–0·08)	9	0·10 (0·04–0·17)
Other or unspecified Gram-negatives	114	0·11 (0·07–0·15)	394	0·12 (0·02–0·25)
Other pathogens	80	0·24 (0·17–0·32)	11	0·01 (0–0·02)

EPIDEMIOLOGIE



The Lancet
Volume 404, Issue 10449, 20–26 July 2024, Pages 277-293

Seminar

Neonatal bacterial sepsis

Tobias Strunk PhD [a b], Eleanor J Molloy PhD [c d e f g], Archita Mishra PhD [b], Zulfiqar A Bhutta PhD [h i]

Fig 2: Mortality trends of neonatal conditions from 1990 to 2019

PROBLEMATIQUES

Diagnostic

- Complétude des fiches de références
- Faible spécificité des signes cliniques
- Biomarqueurs peu performants au cours des 24 premières heures de l'infection
- Négativité de la culture malgré l'infection
- Disponibilité et l'accessibilité des bilans
- Crainte d'une spoliation sanguine sur terrain précaire

PROBLEMATIQUES

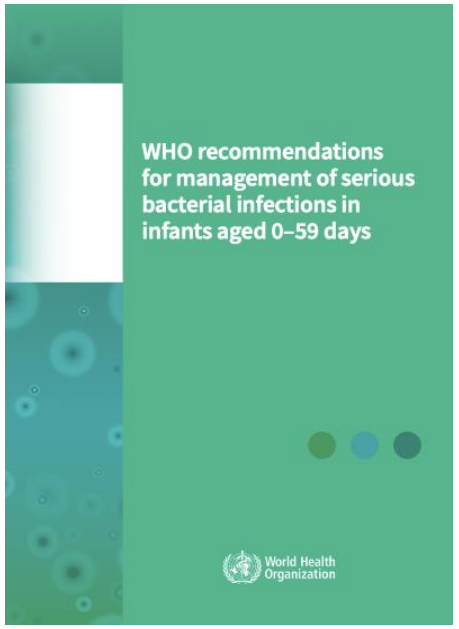
Traitement curatif

❑Qui traiter ?

- Absence de précision sur les FDRI + absence de signes physiques
- Absence de précision sur les FDRI + absence de signes physiques + inaccessibilité des bilans

=> antibiothérapie abusive

Traitement curatif probabiliste OMS 2022



B.2 Suspected sepsis in young infants aged 0–59 days in hospital settings

Ampicillin IM/IV for a total of at least 10 days

- 50 mg/kg/dose every 12 hours (first week of life)
- 50 mg/kg/dose every 8 hours (> first week of life)

Plus

Gentamicin IM/IV for a total of at least 10 days

- 5 mg/kg once a day (first week of life)
- 7.5 mg/kg once a day (> first week of life)

Suspected staphylococcal sepsis in young infants aged 0–59 days in hospital settings

Cloxacillin IM/IV for a total of at least 10 days

- 50 mg/kg/dose every 12 hours (first week of life)
- 50 mg/kg/dose every 8 hours (> first week of life)

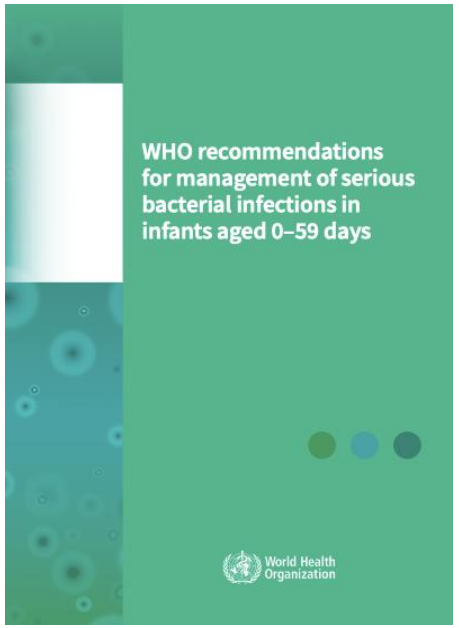
Plus

Gentamicin IM/IV for a total of at least 10 days

- 5 mg/kg once a day (first week of life)
- 7.5 mg/kg once a day (> first week of life)

PROBLEMATIQUES

Curatif probabiliste OMS 2022



B.4 Suspected meningitis in young infants aged 0–59 days in hospital settings

Ampicillin IM/IV for a total of at least 3 weeks

- 50 mg/kg/dose every 12 hours (first week of life)
- 50 mg/kg/dose every 8 hours (> first week of life)

Or

Cefotaxime IM/IV for a total of at least 3 weeks

- 50 mg/kg/dose every 12 hours (first week of life)
- 50 mg/kg/dose every 6 hours (> first week of life)

Or

Ceftriaxone IM/IV for a total of at least 3 weeks

- 100 mg/kg once a day (whether starting in the first week of life or later)

Plus

Gentamicin IM/IV for a total of at least 3 weeks

- 5 mg/kg once a day (first week of life)
- 7.5 mg/kg once a day (> first week of life)

PROBLEMATIQUES

Résistance antimicrobienne

- Gram négatif => 60 % des sepsis néonatals dans les LLIMC
  - Klebsiella spp (38%)+++, 15% E. coli (15%)
- Résistance élevée aux antibiotiques empiriques recommandés par l'OMS
  - Résistances: genta (42-70% tous les gram-); ampi (90% tous les gram-); C3G (88% Klebsiella spp; E. coli 57-81%)

Sophie C. H. Wen[1,2*], Yukiko Ezure[1,3], Lauren Rolley[2]
Gram-negative neonatal sepsis in low- and lower-middle-income countries and WHO empirical antibiotic recommendations: A systematic review and meta-analysis
2021 PLoS Med 18(9): e1003787.

Impact de l'antibiothérapie sur le microbiome

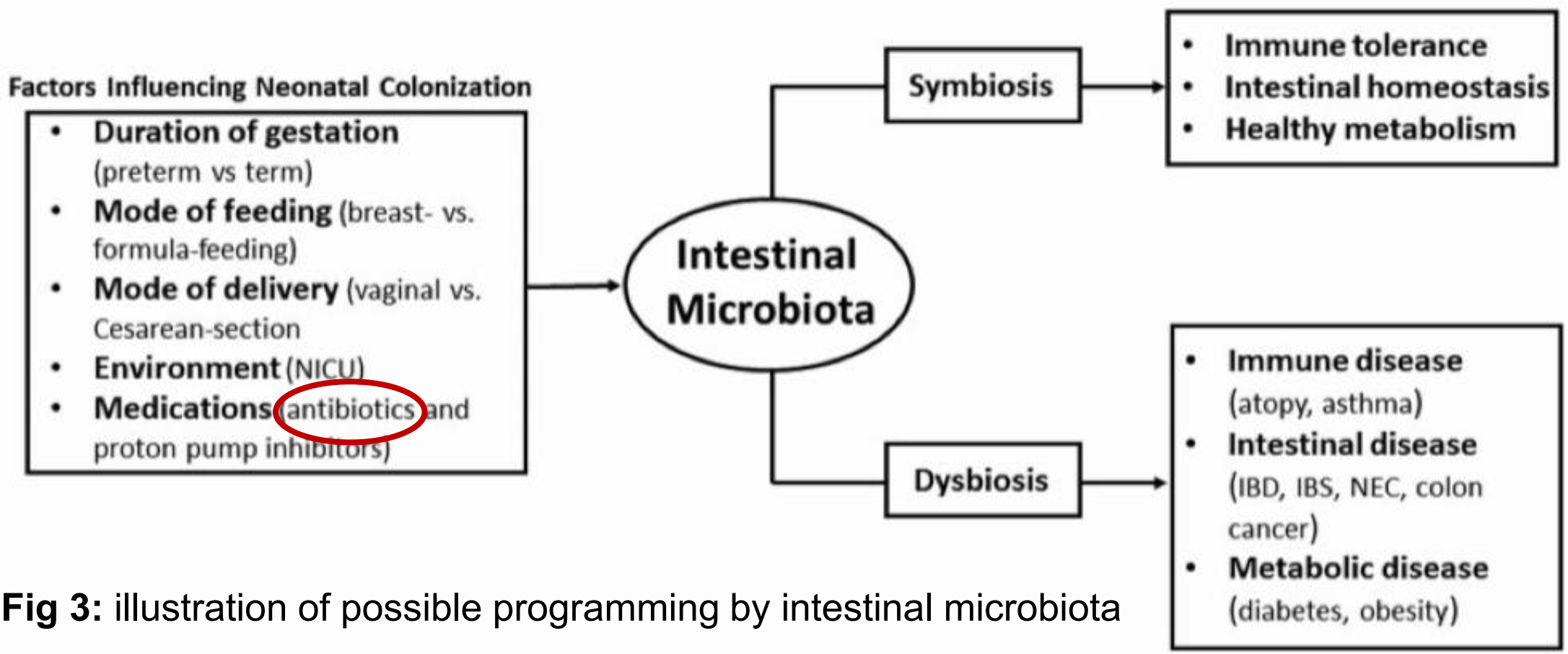


Fig 3: illustration of possible programming by intestinal microbiota

IBD, irritable bowel disease; IBS, irritable bowel syndrome; NEC, necrotizing enterocolitis; NICU, neonatal intensive care unit

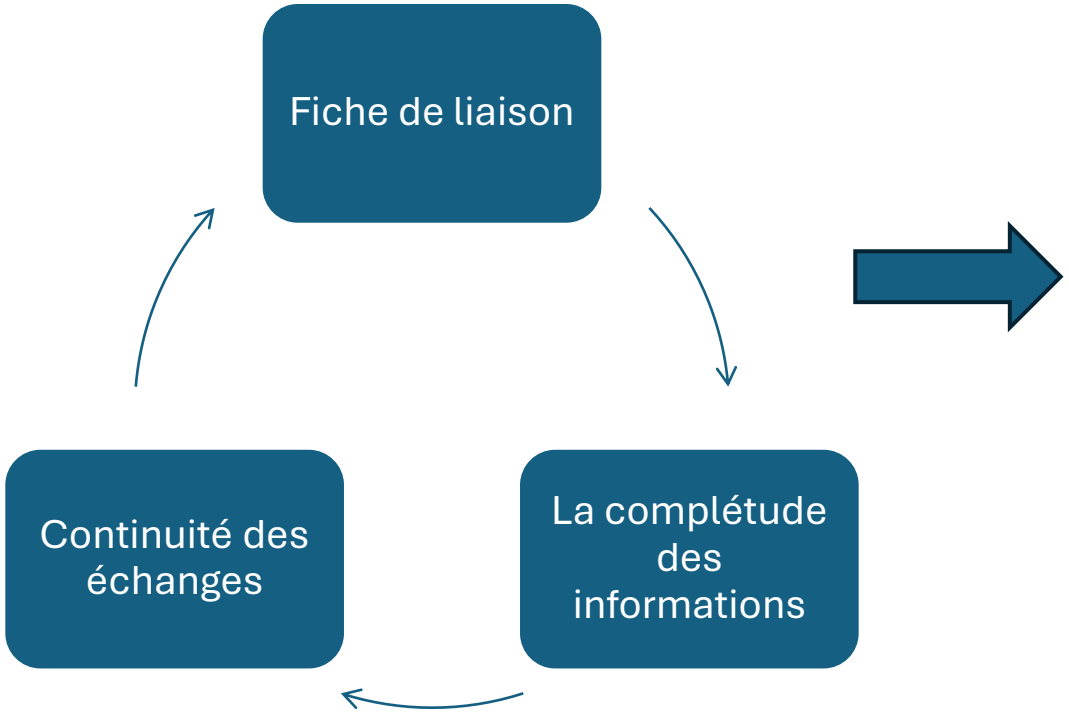
Olivier Goulet

Potential role of the intestinal microbiota in programming health and disease 2015 *Nutrition Reviews® Vol. 73(S1):32–40*

PISTES DE REFLEXION

PISTES DE REFLEXION

Amélioration communication obstétricien-pédiatre sur les FDRI maternels



Prescription rationnelle de l'ATB chez le NN

PISTES DE REFLEXION

Renforcement du continuum de soins, de la période préconceptionnelle et post conceptionnel

- réduirait les infections néonatales de 84 %
- préviendrait près de 2 millions de décès d'ici 2025

Joy E Lawn [1], Hannah Blencowe [2], Shefali Oza

Every Newborn: progress, priorities, and potential beyond survival Lancet. 2014 Jul 12;384(9938):189-205.

PISTES DE REFLEXION

Renforcement du continuum de soins, de la période préconceptionnelle et post conceptionnel

- supplémentation en micronutriments multiples,
- une supplémentation équilibrée en protéines et en énergie,
- l'aspirine à faible dose,
- la progestérone par voie vaginale,
- l'éducation au sevrage tabagique,
- le traitement de la bactériurie asymptomatique
- le traitement de la syphilis,

PISTES DE REFLEXION

Autres mesures de prévention

- Vaccination contre tétanos, coqueluche, H. influenzae
- Vaccin maternelle contre le strep B
- Introduction dans les CPN du dépistage systématique et de la prophylaxie per-partum

Préventif post natal

- Les soins du cordon à la chlorhexidine
- La mise au sein précoce et l'allaitement maternel exclusif
- La surveillance clinique formalisée (Tp°, FC, FR, signes de lutte, coloration) pour une détection précoce des NN infectées et pour l'épargne dans la prescription d'antibiotique
- Utilisation rationnelle des antibiotiques

PISTES DE REFLEXION

Préventif en anté, per et post natal

- Asepsie des mains et des soins
- Surveillance épidémiologique de l'écologie bactérienne et des résistances aux antibiotiques
- Groupes d'expert sous-régionaux africain / réseau => harmonisation des indicateurs, guidelines et de protocole de prise en charge

CONCLUSION

- L'INNB représente un fardeau majeur en Afrique
- Son diagnostic est difficile
- Les streptocoques, les klebsielles et E. coli demeurent les bactéries les plus virulentes
- Résistance élevée des antibiotiques empiriques recommandés par l'OMS
- Bonne collaboration, entre obstétricien et pédiatre

AEROPOR INTERNATIONAL

THANK
YOU

Références

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- **[2]** Fleischmann C, Reichert F, Cassini A, et al. Global incidence and mortality of neonatal sepsis: a systematic review and meta-analysis. Arch Dis Child 2021; 106: 745–52
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